

Predictive Modeling of in-hospitality Mortality Following Elective Surgery:

1. Dataset Summary:

The dataset used in this project is the California inpatient mortality rates and quality ratings. It contains information about various hospitals across California, including mortality rates, number of deaths and cases, procedures/conditions were treated, and the overall hospital quality ratings are categorised as **worse, As Expected, or Better**.

2. Problem Statement:

In-hospital mortality after elective surgery remains a serious concern despite advancements in medical practices and surgical techniques. Early detection of a patient's increased risk of death can lead to better patient survival rates, fewer complications, and timely therapies. However, traditional methods of evaluating risk are often imprecise and fail to recognize complex relationships in patient information. This study intends to develop and evaluate machine learning models that can accurately predict in-hospital mortality using significant clinical and demographic data. The goal of this project is to build machine learning models that can predict a hospital's rating based on its performance metrics. This classification task is important as it can support healthcare quality assessment and potentially help hospitals identify areas for improvement.

3. Predictive Models Implemented:

We implemented four machine learning models to predict the hospital ratings based on different features they are:

- Random Forest Classifier
- XGBoost Classifier
- K-Nearest Neighbours (KNN) classifier
- Support Vector Classifier

Since some hospital rating categories had fewer examples, we balanced the data by adding more samples to the smaller classes using SMOTE, allowing the models to learn better and make fairer predictions. By comparing model performances, we were able to evaluate the impact of class balancing on predictive accuracy.

4. Key Findings and Observations:

- Class imbalances were present in the dataset, with some rating categories were not presented properly. This was addressed using **SMOTE**, which improved the balance of classes in the training set.
- All four models were evaluated using metrics such as **Accuracy, Precision, Recall, F1 Score, ROC-AUC**

- Among all models, Random forest classifier was strongly trained with **SMOTE-balance data** and performed best across multiple metrics, showing some strong classification capability and robustness.
- Balancing the data with SMOTE helped models do a better job in recognising less common hospital ratings, especially improving overall performance across all classes.
- This shows how machine learning can be really useful in evaluating health care quality. Even though this project focuses on predicting hospital ratings, could also be used to predict things like patients risk after surgery in future.